

Dr. Aleksandra Ivanova (formerly Nikonenko)

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Profile

Computational chemist with strong experience in **computer-aided drug design**, **molecular modeling**, **cheminformatics**, and **scientific software development**. My work combines molecular docking, molecular dynamics simulations, binding free-energy analysis, QSAR, *de novo* molecular design, Python-based workflow automation, and HPC execution for small-molecule drug discovery. **Lead developer** of [StreaMD](#) (★100 🗨️21), an open-source toolkit for automated setup, execution, and analysis of molecular dynamics simulations at scale.

Skills

Programming	Python, Bash, R, Cython, SQL, $\text{\LaTeX}$
Drug discovery/CADD	Molecular docking, molecular dynamics simulations, binding free-energy calculations, QSAR <i>de novo</i> design, protein–ligand interaction analysis, automated preprocessing and quality control of chemical datasets
Cheminformatics	RDKit, MDAnalysis, ProLIF, CReM, GROMACS, Amber, AutoDock Vina, Gnina, Glide, Maestro, gmx_MMPBSA, ChemFP, Gaussian, MCPB.py, PyMOL, Chimera
Python ecosystem	scikit-learn, pandas, NumPy, Matplotlib, Plotly, Streamlit
High-performance computing	Dask, PBS, SLURM, multiprocessing, parallel, code performance profiling
Software development	Git/GitHub, package distribution (PyPI, conda-forge), environment management (conda, venv), containerization (Docker, Apptainer), REST API design, construction of reproducible workflows, unit testing, integration testing

Experience

2026–	Researcher Institute of Molecular and Translational Medicine, Palacký University, Czech Republic Conduct computational drug discovery research involving molecular dynamics simulations, molecular docking, binding free-energy analysis, QSAR modeling, and <i>de novo</i> molecular design. Develop Python-based workflows and open-source tools for reproducible molecular modeling, including StreaMD and CReM-opt.
2022/06–12	Computer Scientist (part-time) Allffinity s.r.o, Czech Republic Curated and validated protein structure (PDB) datasets for MD and ML pipelines; performed molecular docking using Glide.
2017–2019	Laboratory Assistant Laboratory of Chemistry and Biochemistry, Volgograd State University, Russia Provided laboratory assistance in chemistry and biochemistry, including sample preparation, handling laboratory equipment, maintaining laboratory materials, and supporting routine experimental procedures.

Education

2019–2026	Ph.D., Molecular and Translational Medicine — Computational Chemistry Palacký University, Czech Republic Thesis: <i>From molecular conformation to biological activity: computational methods for the design of small molecules</i> . Supervisor: Dr. Pavel Polishchuk
2014–2019	Specialist (Honours), Bioengineering & Bioinformatics Volgograd State University, Russia

Projects

Lead Developer

2022–	StreaMD ★100 🗨️21  GitHub Developed an open-source toolkit for automated setup, execution, and analysis of high-throughput molecular dynamics simulations. Published in <i>J. Cheminformatics</i> (2024). <i>First-author publication</i> .
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- 2021–
CReM-opt  [GitHub](#)
Developed a docking-guided molecular optimization framework combining CReM fragment replacement, genetic algorithms, and atom-wise docking score attribution. Presented at *RDKit UGM 2025*. Poster Award at *12th ICCS 2022, Netherlands*. *Manuscript in preparation*.
- 2019–2022
MIL-based QSAR Modeling
Developed a conformer-based multi-instance learning approach for molecular activity prediction. Automated the collection of activity-cliff datasets (stereoisomers with differential biological activity). *First-author publication*. Contributed to a related study through dataset curation and molecular docking.
- 2019
ChEMBL Datasets Collection  [GitHub](#)
Developed an automated pipeline for retrieving and preprocessing bioactivity datasets from ChEMBL, tailored to specific activity types (e.g., agonists, antagonists, inverse agonists).
- 2025
HPC Stats Scripts  [GitHub](#)
Developed utilities for monitoring SLURM/PBS job efficiency and CPU/GPU/memory usage in HPC environments.

Selected Research Contributions

- 2023
CACHE Challenge #1 — 3rd Place Team WDR Domain of LRRK2
Developed and applied a docking-guided *de novo* molecule generation workflow using CReM-opt and performed MD-based evaluation of identified hits.
- 2021–2025
Tubulin Inhibitor SAR Studies co-author of 4 publications
Performed molecular docking, molecular dynamics simulations, free energy calculations, ProLIF-based protein–ligand interaction analysis, parameterization of boron-containing molecules using Gaussian.
- 2022
EasyDock  [GitHub](#)
Implemented a Python-based AutoDock Vina workflow as part of a scalable molecular docking tool.

Selected Publications

9 total — [full list on ORCID](#)

- A. **Ivanova**, O. Mokshyna, and P. Polishchuk. “StreaMD: the toolkit for high-throughput molecular dynamics simulations”. In: *Journal of Cheminformatics* (2024). DOI: 10.1186/s13321-024-00918-w
- A. **Nikonenko** et al. “Multiple Conformer Descriptors for QSAR Modeling”. In: *Molecular Informatics* (2021). DOI: 10.1002/minf.202060030
- F. Li et al. (incl. **Ivanova** A.). “CACHE Challenge #1: Targeting the WDR Domain of LRRK2, A Parkinson’s Disease Associated Protein”. In: *Journal of Chemical Information and Modeling* (2024). DOI: 10.1021/acs.jcim.4c01267

Honors & Awards

- 2026
3rd Place (individual on-site entry) *in silico* Drug Design Challenge
Workshop “Advanced *in silico* Drug Design”
- 2025
Dean’s Award Palacký University
Significant publication activity 2024 — “StreaMD: the toolkit for high-throughput molecular dynamics simulations”
- 2023
3rd Place Team CACHE Challenge #1
Prediction and experimental validation of hits for WDR domain of LRRK2
- 2022
Poster Award 12th ICCS, Netherlands
GenCReM: *de novo* molecular generation
- 2022
Dean’s Award Palacký University
Significant publication activity 2021 — “Multiple Conformer descriptors for QSAR modeling”

Lectures & Workshops

- 2023–2026
Lecturer, annual workshop “Advanced *in silico* Drug Design”
Palacký University, Olomouc, Czech Republic
Molecular Dynamics Workshop
- 2025
Talk, RDKit UGM 2025 Prague, Czech Republic
CReM-opt: *de novo* design guided by explainable docking